PRANESH U S

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Education

Bachelor of Technology in Computer Science and Business Systems *2024 – 2028*
Sri Krishna College of Engineering and Technology CGPA: 8.21

Experience

Software Engineering Intern
Infosys Springboard Jan 2026 – Mar 2026

- Fortified MedVault, a healthcare platform, by integrating Spring Security, JWT, and RBAC to secure REST APIs and MySQL databases.
- Architected a refresh-token authentication flow with blacklist-based logout, reducing unauthorized access vulnerabilities by 90%.
- Streamlined secure patient record retrieval by designing normalized MySQL schemas, cutting query latency by 30% for 10,000+ records.

Java Development Intern
Pinnacle Lab (Remote) Jun 2025 – Aug 2025

- Architected Flight Booking and Library Management systems utilizing advanced Java OOP principles, improving code reusability by 40%.
- Deployed SeedCart, a Spring Boot backend automating dynamic database seeding for 50+ test scenarios, cutting setup time by 60%.
- Formulated layered backend architectures and optimized REST API endpoints, ensuring scalable data handling for high-traffic modules.

Projects

VPC-Native AI Gateway — Secure BPO Infrastructure *2026*
Java · Spring Boot · React.js · Redis · Kafka · Presidio

- Engineered a secure, VPC-native AI gateway for BPOs, integrating Microsoft Presidio for task-aware PII masking and Shannon Entropy-based smart model routing.
- Orchestrated a SHA-256 blockchain-style audit ledger and asynchronous Kafka streaming to ensure tamper-proof DPDP Act compliance.
- Constructed a real-time Redis micro-billing ledger and dual-filter security scanner, reducing AI API costs by 40% while preventing hallucinated data leaks.

The Debugger Tool — Automated Error Resolution & AST Parsing *2026*
Python · React.js · AST Parsing · Regex

- Engineered a robust debugging assistant that parses complex stack traces and source code using Abstract Syntax Tree (AST) analysis to accurately pinpoint root causes.
- Developed an automated error-mapping algorithm to trace exception propagation across multiple files, reducing manual debugging time by 45%.
- Crafted an interactive React.js UI for developers to visualize error propagation trees and apply targeted code patches, accelerating the development lifecycle.

AuthForge & SecureBlog — Security Frameworks *2025-2026*
Java · Spring Boot · Spring Security · React.js · PostgreSQL

- Formulated AuthForge: a reusable Spring Security authentication framework supporting JWT, access/refresh tokens, and custom filters, reducing integration time by 50%.
- Launched SecureBlog: a full-stack platform featuring role-based access control, rich text editing, and real-time search, handling 500+ concurrent user simulations.
- Optimized PostgreSQL database schemas and reusable authentication modules, enabling scalable, secure application development with minimal configuration.

Technical Skills

- **Languages:** Java, JavaScript, Python, SQL, HTML, CSS
- **Frameworks:** Spring Boot, Spring Security, React.js, Tailwind CSS, FastAPI
- **Backend/Infra:** REST APIs, JWT, OAuth2, Redis, Apache Kafka, Docker, VPC Architecture, SHA-256
- **Database/Cloud:** MySQL, PostgreSQL, Firebase, AWS (EC2, IAM, S3), Git, GitHub, Postman

Certifications

- **Claude Code in Action** (2026)
- **Intro to LLMs** - Google Cloud (May 2025)
- **Python for Data Science** - IBM (Aug 2025)
- **Software Eng. / Intro to AI / DB & SQL** - Infosys (2026)
- **Networking Basics** - Cisco (Jan 2025)

Data Structures & Algorithms

- Strong grasp of arrays, linked lists, trees, graphs, hash maps, recursion, and dynamic programming, with regular practice in Java emphasizing time/space complexity.
- Applied DSA concepts directly to backend development via optimized MySQL queries, efficient data retrieval, and scalable REST API design.